Paul Downing

Senior Software Engineer | React, TypeScript, Ruby on Rails

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SUMMARY

Senior Software Engineer with 10+ years writing production and test code, including 9 years building web applications with React, TypeScript, and Ruby on Rails. Spent 5 years owning the frontend of a DeFi platform that processed millions of transactions; currently the founder and sole engineer of a live Rails and React sports prediction platform with Python models. Strong in frontend architecture, complex transaction workflows, test automation, release coordination, and mentoring engineers on remote teams. Uses AI tools like Claude Code to speed up development and data research, and reviews and owns every line that ships.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Ruby, Python, HTML, CSS

Frontend: React, Redux, Inertia.js, Vite, Webpack, Babel, Material UI (MUI), Tailwind CSS, Bootstrap, styled-components

Backend: Ruby on Rails, ActiveRecord, REST APIs, Node.js

Testing: RSpec, Minitest, Jest, React Testing Library, Playwright, Cypress, Capybara, Cucumber, Selenium WebDriver

Web3: MetaMask, Keplr, Ledger, and Trust Wallet integration, EVM, Cosmos, IBC, cross-chain bridging

Tools and Delivery: Git, GitHub, GitLab, CI/CD, RuboCop, Prettier, Heroku, Netlify, Docker, AWS (S3, EC2), npm, Sentry, Google Analytics, A/B testing, Figma, Jira, Shortcut, Agile/Scrum

AI: Claude Code, OpenAI Codex, AI-assisted development (coding agents), AI-assisted data research for prediction models

EXPERIENCE

Founder and Software Engineer 01/2026 - Present
Paragon Sports Intelligence, Denver, CO (Remote)

- Built and launched paragon-sports-intelligence.com, an AI-powered sports prediction platform, as the sole engineer: Ruby on Rails backend and REST API, React and TypeScript frontend using Inertia.js, deployed on Heroku; ~40 daily active users.
- Developed Python prediction models for NCAAM games (~80% accuracy picking winners, ~7-point mean absolute error), MLB scores and starting pitchers, and NFL scores and player props (pre-release), using AI tools to accelerate data research.
- Built a CI/CD pipeline that runs RSpec and React Testing Library suites plus RuboCop and Prettier checks, and use AI tools (Claude Code, OpenAI Codex) to speed up delivery.

Software Engineering Manager 10/2025 - 12/2025
Kava Labs, Remote

Kava Labs builds Kava, a DeFi blockchain platform (app.kava.io) for minting, lending, swapping, and cross-chain bridging, with $2.5B+ in bridged asset transactions.

- Promoted to a player-coach manager role, splitting time 50/50 between hands-on frontend development and leading 3 engineers through 1:1s, performance reviews, and growth plans, formalizing work led informally since 2023.

Senior Software Engineer II 10/2020 - 10/2025
Kava Labs, Remote

- Owned frontend architecture for the Kava web app (React, Redux, TypeScript, Material UI) across the Mint (CDP), Lend, Swap, and Bridge products, which processed millions of transactions.
- Led the frontend of a 6-month cross-chain bridge project moving assets between Kava's EVM and Cosmos chains and to partner Cosmos chains via IBC, building precise transaction messaging, balance and system checks, and success and failure workflows.
- Integrated MetaMask, Keplr, Ledger, and Trust Wallet, and engineered Kava's JavaScript SDK npm package; contributed to third-party crypto packages.
- Served as frontend release coordinator for quarterly network upgrades, carrying releases through testnet validation and on-chain governance voting.

- Planned frontend delivery with product: mapped user stories, wrote and sequenced tickets and timelines, ran technical spikes to choose approaches, assessed risk, and managed frontend sprints.
- Partnered with UX on user workflows and wireframes; built A/B testing and Google Analytics event tracking to measure conversion.
- Created the company's bug workflow (user-impact and severity assessment, biweekly on-call handoff, clear ownership) to shorten bug cycle time, and served as the primary escalation point for production frontend issues, earning the internal nickname "Better Call Paul."
- Helped implement a CI/CD pipeline that ran every test suite (Jest, Cypress, Playwright) before deployment, reducing regressions, and maintained 80%+ test coverage.
- Led the conversion of the codebase from JavaScript to TypeScript, and set up Sentry error monitoring with Slack alerts to speed up production bug fixes.
- Onboarded and mentored 3 engineers in React, TypeScript, and build tooling, all of whom were later promoted; wrote a new-hire onboarding guide that cut technical setup time by days, and designed a React pair-programming exercise used in frontend hiring.

Software Engineer 10/2019 - 04/2020
TeamSnap, Remote

TeamSnap builds a sports team management app for coaches, players, and families.

- Helped rebuild the account management interface, redesigning data models and integrating multiple REST API endpoints (project halted by the 2020 COVID shutdown).
- Migrated the account management application from MobX to Redux and Redux Form, aligning it with 4 other company frontend projects so more engineers could support it.
- Raised code coverage from 9% to 78% on a medium-sized production application, and added unit tests and CI/CD pipelines to 3 projects.
- Built a shared design library with product and UX, standardizing components across 6 frontend applications; built client registration forms in Ruby on Rails and jQuery.

Software Engineer 01/2017 - 10/2019
Granicus, Remote

Granicus builds communications and meeting management software for government agencies.

- Rewrote a user-facing application from Ember.js to React, using Redux for state management and Axios against a Ruby on Rails API.
- Introduced Jest testing patterns for React applications, raising frontend code coverage from 30% to 70%.
- Built Ruby on Rails features and mentored a colleague with no coding background until they could contribute to the Rails codebase.
- Developed features for real-time meeting minutes software using WebSockets.
- Contributed to a WYSIWYG drag-and-drop email builder for creating templated emails.

Software Automation Engineer 10/2015 - 01/2017
Granicus, Saint Paul, MN

Hired at GovDelivery, which merged with Granicus in 2016.

- Automated 1,500+ test cases in Ruby: 1,200+ covering new features and 300+ backlog cases that had been run manually every release.
- Helped cut the release regression cycle from ~10 days to 1 day by converting manual backlog tests into automated integration, controller, and model tests.
- Wrote UI tests in Cucumber, Capybara, and Selenium WebDriver; supported development and QA for datacenter failover deployments.

EDUCATION AND TRAINING

Bachelor of Science, Business and Sports Management 05/2012
University of Minnesota, Minneapolis, MN

Training: Ruby on Rails Tutorial (Michael Hartl), Udemy programming coursework, UX design workshop